

SyedAhsan_Sirat_Meal_Planner.ipynb

Setup: Phase 1

Installing required libraries

```
# Install the required libraries
%pip install recipe-scrappers termcolor tabulate
```

Requirement already satisfied: recipe-scrappers in /usr/local/lib/python3.11/dist-packages (15.4.0)
Requirement already satisfied: termcolor in /usr/local/lib/python3.11/dist-packages (2.5.0)
Requirement already satisfied: tabulate in /usr/local/lib/python3.11/dist-packages (0.9.0)
Requirement already satisfied: beautifulsoup4>=4.12.3 in /usr/local/lib/python3.11/dist-packages (from recipe-scrappers) (4.13.3)
Requirement already satisfied: extract>=0.17.0 in /usr/local/lib/python3.11/dist-packages (from recipe-scrappers) (0.18.0)
Requirement already satisfied: isodate>=0.6.1 in /usr/local/lib/python3.11/dist-packages (from recipe-scrappers) (0.7.2)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.11/dist-packages (from beautifulsoup4>=4.12.3->recipe-scrappers) (0.18.0)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.11/dist-packages (from beautifulsoup4>=4.12.3->recipe-scrappers) (4.12.3)
Requirement already satisfied: lxml in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (5.3.1)
Requirement already satisfied: lxml-html-clean in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (0.4.1)
Requirement already satisfied: rdflib>=6.0.0 in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (7.1.3)
Requirement already satisfied: pyrdfa3 in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (3.6.4)
Requirement already satisfied: mf2py in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (2.0.1)
Requirement already satisfied: w3lib in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (2.3.1)
Requirement already satisfied: html-text>=0.5.1 in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (0.7.1)
Requirement already satisfied: jstyleson in /usr/local/lib/python3.11/dist-packages (from extract>=0.17.0->recipe-scrappers) (0.0.2)
Requirement already satisfied: pyparsing<4,>=2.1.0 in /usr/local/lib/python3.11/dist-packages (from rdflib>=6.0.0->extract>=0.17.0->recipe-scrappers) (3.1.2)
Requirement already satisfied: html5lib<2.0,>=1.1 in /usr/local/lib/python3.11/dist-packages (from mf2py->extract>=0.17.0->recipe-scrappers) (1.1.0)
Requirement already satisfied: requests<3.0.0,>=2.28.2 in /usr/local/lib/python3.11/dist-packages (from mf2py->extract>=0.17.0->recipe-scrappers) (2.32.0)
Requirement already satisfied: six>=1.9 in /usr/local/lib/python3.11/dist-packages (from html5lib<2.0,>=1.1->mf2py->extract>=0.17.0->recipe-scrappers) (1.17.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.11/dist-packages (from html5lib<2.0,>=1.1->mf2py->extract>=0.17.0->recipe-scrappers) (0.11)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.28.2->mf2py->extract>=0.17.0->recipe-scrappers) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.28.2->mf2py->extract>=0.17.0->recipe-scrappers) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.28.2->mf2py->extract>=0.17.0->recipe-scrappers) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.28.2->mf2py->extract>=0.17.0->recipe-scrappers) (2025.1.1)

Setup: Phase 2

Testing the libraries

```
from urllib.request import urlopen
from recipe_scrappers import scrape_html
from termcolor import colored
from collections import Counter
from tabulate import tabulate
import statistics

url = 'https://www.allrecipes.com/recipe/25189/chicken-alfredo-pizza/'
try:
    # Retrieve the recipe webpage HTML
    html = urlopen(url).read().decode("utf-8")
    # Pass the HTML and URL to scrape_html
    scraper = scrape_html(html, org_url=url)
    # Extract and display recipe information
    print("Title:", scraper.title())
    print("Ingredients:", scraper.ingredients())
    print("Total Time:", scraper.total_time())
    print("Yields:", scraper.yields())
    print("Instructions:", scraper.instructions())
    print(colored("Test Complete", "green"))

except ImportError as e:
    print(colored("Error: Unable to import scrape_html.", "red"))
except Exception as e:
    print(colored("An error occurred: " + str(e), "red"))
```

```
Title: Chicken Alfredo Pizza
Ingredients: ['2 tablespoons butter', '1 clove garlic, minced', '1 pinch dried rosemary', '1 pinch salt', '4 tablespoons butter', '4 tab
Total Time: 100
Yields: 8 servings
Instructions: Make garlic butter: Melt butter in a small saucepan over medium heat. Blend in garlic, rosemary, and salt. Cook and stir u
Make sauce: Melt butter in a small saucepan over medium heat. Blend flour, salt, and pepper into melted butter. Stir in milk and Romano
Make dough: Stir yeast into warm water in a small bowl until yeast is dissolved. Allow to rest until yeast foams, about 5 minutes. Mix t
```

Preheat the oven to 400 degrees F (200 degrees C).

Prepare chicken: Season roasted chicken with rosemary, thyme, poultry seasoning, garlic powder, and salt. Chop or shred chicken and set

Assemble pizza: Spread dough out onto a pizza pan. Brush on garlic butter, covering entire crust. Spread sauce on top, leaving crust edge

Bake in the preheated oven until the bottom crust is lightly browned, about 20 minutes. Allow pizza to rest for 2 to 3 minutes before sl

Test Complete

Assignment Task

✓ Step 1: Scrape Recipes

Using recipe-scrappers to scrape five recipes. Storing the scraped data in a list.

```
from recipe_scrappers import scrape_html
from urllib.request import urlopen
#using this recipes list to store receipes
recipes = []
meal_plan=[]
# Example recipe scraping
#Using a list of url for 5 recipes
url = []
url.append( 'https://www.allrecipes.com/slow-cooker-chicken-enchilada-casserole-recipe-8788263')
url.append( 'https://www.allrecipes.com/air-fried-tossed-taquitos-recipe-8784154')
url.append( 'https://www.allrecipes.com/taquito-casserole-recipe-8673259')
url.append( 'https://www.allrecipes.com/creamy-mississippi-chicken-enchiladas-recipe-8678116')
url.append( 'https://www.allrecipes.com/air-fryer-chicken-quesadillas-recipe-8651412')
i=0
while i<5:
    html = urlopen(url[i]).read().decode("utf-8")
    scraper = scrape_html(html, org_url=url[i])
    recipes.append({
        "title": scraper.title(),
        "ingredients": scraper.ingredients(),
        "instructions": scraper.instructions()
    })
    print("Recipes loaded!")
    i+=1
```

Recipes loaded!
 Recipes loaded!
 Recipes loaded!
 Recipes loaded!
 Recipes loaded!

✓ Step 2: Special Recipes

Adding a Bangladeshi style Chicken Vuna (Custom) recipe

```
chicken_vuna_recipe = {
    'title': 'Simple Chicken Vuna',
    'ingredients': [
        '1 pound chicken, cut into pieces',
        '2 tablespoons oil',
        '1 onion, chopped',
        '1/2 teaspoon turmeric',
        '1/2 teaspoon salt'
    ],
    'instructions': (
        'Heat oil in a pan, sauté onions until soft.\n'
        'Add chicken, turmeric, and salt. Cook until tender.\n'
        'Serve with rice or roti.'
    )
}

recipes.append(chicken_vuna_recipe)
```

✓ Step 3: Menu and User Interaction

Creating Menu for user

```
# Printing the Menu
print("Menu: Enter a number to navigate the menu")
print("""
1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit""")

# Getting input from user
optionSelected = int(input("Enter a number: "))

#Using a loop to navigate the menu
while(optionSelected != 6):
    if optionSelected == 1:

        print("\nAvailable Recipes:")
        for i in range(len(recipes)):
            print(f"{i+1}. {recipes[i]['title']}")

    elif optionSelected ==2:
        # View available recipes
        print("\nAvailable Recipes:")
        for i in range(len(recipes)):
            print(f"{i+1}. {recipes[i]['title']}")

        # Ask user to select a recipe to add to the meal plan
        choice = int(input("Select a number to add to your meal plan: "))

        if 1 <= choice <= len(recipes):
            selected_recipe = recipes[choice - 1]
            meal_plan.append(selected_recipe)
            print(f"{selected_recipe['title']} added to your meal plan!")
        else:
            print("Invalid choice.")

    elif optionSelected ==3:
        # View meal plan (list of recipes added)
        if len(meal_plan)==0:
            print("\nYour meal plan is empty.")
        else:
            print("\nYour Meal Plan:")
            for i in range(len(meal_plan)):
                print(f"{i+1}. {meal_plan[i]['title']}")

            # Ask user to select a recipe to view full details
            choice = int(input("Select a number to view the full Meal Plan OR Selec 0 to go back: "))
            # Display selected recipe details
            if choice==0:
                print("Going Back...")
            elif 1 <= choice <= len(meal_plan):
                selected_meal = meal_plan[choice - 1]
                print(f"\nFull Recipe: {selected_meal['title']}")
                print("Ingredients:")
                for ingredient in selected_meal['ingredients']:
                    print(f"- {ingredient}")
                print("\nInstructions:")
                print(selected_meal['instructions'])
            else:
                print("Invalid choice. Please select a valid number.")

    elif optionSelected ==4:
        # Generate shopping list from meal plan
        shopping_list = Counter()
        for meal in meal_plan:
            for ingredient in meal["ingredients"]:
                shopping_list[ingredient] += 1
```

```

# Display shopping list in a table format
print("\nShopping List:")
print(tabulate(shopping_list.items(), headers=["Ingredient", "Quantity"], tablefmt="grid"))

#Color Shopping List
"""
Note: I tried to use this code found in assignment but it is not supported by google colab

from termcolor import colored
print(colored("Shopping List", "blue"))
for item, qty in shopping_list.items():
    print(colored(item, "green"), ":", colored(qty, "blue"))
"""

from IPython.display import display, HTML

print(colored("Shopping List in color format", "blue"))

# Loop through the shopping list and print each item with colored text
for item, qty in shopping_list.items():
    # Create colored HTML text for each item and quantity
    item_colored = f'<span style="color:green;">{item}</span>'
    qty_colored = f'<span style="color:blue;">{qty}</span>'

    # Display the colored item and quantity
    display(HTML(f'{item_colored} : {qty_colored}'))

elif optionSelected ==5:
    ingredient_counts = [len(recipe["ingredients"]) for recipe in meal_plan]
    if ingredient_counts:
        print("Average ingredients:", statistics.mean(ingredient_counts))
        print("Median ingredients:", statistics.median(ingredient_counts))
    else:
        print("No recipes in meal plan to calculate statistics.")
else:
    print("Select an option from (1/2/3/4/0)")
#If-Else Ends here

# Printing the Menu
print('.....')
print("Menu: Enter a number to navigate the menu")
print("""
1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit""")

# Getting input from user
optionSelected = int(input("Enter a number: "))

```

↩ Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit

Enter a number: 1

Available Recipes:

1. Slow Cooker Chicken Enchilada Casserole
2. Air Fried Tossed Taquitos
3. Taquito Casserole
4. Creamy Mississippi Chicken Enchiladas
5. Air Fryer Chicken Quesadillas
6. Simple Chicken Vuna

.....

Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit

Enter a number: 3

Your meal plan is empty.

.....

Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit

Enter a number: 2

Available Recipes:

1. Slow Cooker Chicken Enchilada Casserole
2. Air Fried Tossed Taquitos
3. Taquito Casserole
4. Creamy Mississippi Chicken Enchiladas
5. Air Fryer Chicken Quesadillas
6. Simple Chicken Vuna

Select a number to add to your meal plan: 1

Slow Cooker Chicken Enchilada Casserole added to your meal plan!

.....

Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit

Enter a number: 3

Your Meal Plan:

1. Slow Cooker Chicken Enchilada Casserole

Select a number to view the full Meal Plan OR Selec 0 to go back: 0

Going Back...

.....

Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit

Enter a number: 2

Available Recipes:

1. Slow Cooker Chicken Enchilada Casserole
2. Air Fried Tossed Taquitos
3. Taquito Casserole
4. Creamy Mississippi Chicken Enchiladas
5. Air Fryer Chicken Quesadillas
6. Simple Chicken Vuna

Select a number to add to your meal plan: 6

Simple Chicken Vuna added to your meal plan!

```

.....
Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit
Enter a number: 2

Available Recipes:
1. Slow Cooker Chicken Enchilada Casserole
2. Air Fried Tossed Taquitos
3. Taquito Casserole
4. Creamy Mississippi Chicken Enchiladas
5. Air Fryer Chicken Quesadillas
6. Simple Chicken Vuna
Select a number to add to your meal plan: 4
Creamy Mississippi Chicken Enchiladas added to your meal plan!
.....
Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit
Enter a number: 3

Your Meal Plan:
1. Slow Cooker Chicken Enchilada Casserole
2. Simple Chicken Vuna
3. Creamy Mississippi Chicken Enchiladas
Select a number to view the full Meal Plan OR Selec 0 to go back: 4
Invalid choice. Please select a valid number.
.....
Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit
Enter a number: 4

Shopping List:
+-----+-----+
| Ingredient | Quantity |
+-----+-----+
| 1 pound skinless boneless chicken thighs | 1 |
+-----+-----+
| 1 (15-ounce) can black beans | 1 |
+-----+-----+
| 1 cup corn | 1 |
+-----+-----+
| 1/2 cup onion | 1 |
+-----+-----+
| 2 cloves garlic, | 1 |
+-----+-----+
| 1/2 teaspoon salt | 2 |
+-----+-----+
| 1/2 teaspoon ground black pepper | 1 |
+-----+-----+
| 1/2 teaspoon ground cumin | 1 |
+-----+-----+
| 2 (10-ounce) cans enchilada sauce | 1 |
+-----+-----+
| 1 (8-ounce) package Mexican blend cheese, | 1 |
+-----+-----+
| 8 (6-inch) corn tortillas | 1 |
+-----+-----+
| iceberg or Romaine lettuce | 1 |
+-----+-----+
| fresh tomato | 1 |
+-----+-----+
| chopped green onion | 1 |
+-----+-----+
| sour cream | 1 |
+-----+-----+
| 1 pound chicken, cut into pieces | 1 |
+-----+-----+
| 2 tablespoons oil | 1 |

```

1 onion, chopped	1
1/2 teaspoon turmeric	1
2 pounds boneless chicken breasts	1
1 (1 ounce) package dry ranch dressing mix	1
1 cup pepperoncini peppers	1
1/4 cup pepper juice	1
4 tablespoons butter	1
1/4 cup water	1
8 ounces cream cheese	1
1/2 cup sour cream	1
3 cups shredded Monterey Jack cheese	1
8 (10 inch) flour tortillas	1
1 1/2 cups heavy cream	1
sliced green onions	1

Shopping List in color format

1 pound skinless boneless chicken thighs : 1
1 (15-ounce) can black beans : 1
1 cup corn : 1
1/2 cup onion : 1
2 cloves garlic, : 1
1/2 teaspoon salt : 2
1/2 teaspoon ground black pepper : 1
1/2 teaspoon ground cumin : 1
2 (10-ounce) cans enchilada sauce : 1
1 (8-ounce) package Mexican blend cheese, : 1
8 (6-inch) corn tortillas : 1
iceberg or Romaine lettuce : 1
fresh tomato : 1
chopped green onion : 1
sour cream : 1
1 pound chicken, cut into pieces : 1
2 tablespoons oil : 1
1 onion, chopped : 1
1/2 teaspoon turmeric : 1
2 pounds boneless chicken breasts : 1
1 (1 ounce) package dry ranch dressing mix : 1
1 cup pepperoncini peppers : 1
1/4 cup pepper juice : 1
4 tablespoons butter : 1
1/4 cup water : 1
8 ounces cream cheese : 1
1/2 cup sour cream : 1
3 cups shredded Monterey Jack cheese : 1
8 (10 inch) flour tortillas : 1
1 1/2 cups heavy cream : 1
sliced green onions : 1

.....
Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan
4. Generate Shopping List
5. Calculate Statistics
6. Exit

Enter a number: 5

Average ingredients: 10.666666666666666

Median ingredients: 12

.....
Menu: Enter a number to navigate the menu

1. View Recipes
2. Add Recipe to Meal Plan
3. View Meal Plan